

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

19th July 2022

Subject CP3 – Communication Practice

Time allowed: 3 Hours 30 Minutes (09.30 – 13.00 Hours)

Total Marks: 100

You have recently joined a mid-size General Insurance Company as the Appointed Actuary. The Company has been predominantly writing Motor Insurance business for two decades. The Company also has a small portfolio of Commercial Insurance products and it started writing Health Insurance business few years ago. In recent times, the Company has struggled to achieve the intended business growth which has not been looked at favourably by the Board of Directors. On top of that the profitability of the Company has also declined.

You are about to attend your first Board Meeting in this Company and as preparation, you had one briefing session with the CEO of the Company. During the discussion, the CEO has shared the views from the Directors on impact of Company's reserving practices towards the profitability. He has also shared a part of an email written by one of the Directors to him expressing some concern. The same is given below:

“While it is heartening to see some of the decisions on business growth have started delivering results, I am having concern about our overall claims reserving practices. I feel we are very conservative which has directly affected the profitability of last financial year.

I was informed by the CFO that we have large volume of business in Motor Third Party liability and a significant proportion is from Zone Z, hence reserving needs to be prudent. But I am aware that many of our competitors also have similar business, whereas they are at a much better profitability position. It was also told to me that our Actuaries calculate reserves by different methods and consider the most prudent number out of those.

There has been significant improvement in the road conditions in our country which means we could expect lesser number of accidents than past. Why are we not experiencing any benefit out of this on our P&L?

It would be helpful if the new Appointed Actuary can explain me some of these aspects on claims reserving before the Board Meeting.”

The Director has joined the Board last year only and does not come from general insurance background.

- Q. 1)** Draft a note in response to the email of the Director explaining in simple language the reserving approach, especially for Motor Third Party Liability business by addressing the specific concern areas raised in his email.

The Reserving Actuary has shared with you various supporting materials for drafting your response.

[90]

- Q. 2)** State with reasons, four terms which you considered as jargon and left out of your reply or explained so that the Director can understand them?

[4]

- Q. 3)** One of the ways that communication with non-technical audience can be made effective is by leaving out information which may not be relevant for the audience. Explain what information you considered as irrelevant and left out of your reply.

[6]

A) Supporting materials shared by the Reserving Actuary

- 75% of the Company's Gross Written Premium comes from Motor, while 15% and 10% are from Commercial and Accident & Health lines of business respectively.
- 90% of the claims reserve (outstanding claims reserve + Reserve for Incurred But Not Reported claims) pertains to Motor Third Party (TP) Liability.
- Company's Motor book has higher proportion in Private Car, whereas the growth in Commercial Vehicle categories has picked up in last three years, Two-wheeler forms a relatively insignificant part.

GWP Mix	FY 2017-18	FY 2018-19	FY 2019-20	FY 2020-21	FY 2021-22
Private Car	75%	74%	70%	67%	63%
Commercial Vehicle	20%	22%	25%	29%	34%
Two-wheeler	5%	4%	5%	4%	3%

- Within the Commercial Vehicle segment, majority of the business is written under higher vehicle weight category where the expected losses are higher. This was due to certain management action towards targeted growth.

Gross Vehicle Weight	GWP mix
Not exceeding 7500 kg	5%
Exceeding 7500 kg but not exceeding 12000 kg	20%
Exceeding 12000 kg but not exceeding 20000 kg	35%
Exceeding 20000 kg	40%

- The latest Ultimate Loss Ratio estimates are as follows:

Segment	ULR estimate
Private Car	65%
Commercial Vehicle	
Not exceeding 7500 kg	55%
Exceeding 7500 kg but not exceeding 12000 kg	75%
Exceeding 12000 kg but not exceeding 20000 kg	90%
Exceeding 20000 kg	120%
Two-wheeler	70%

- IBNR estimation methods for Motor TP liability:
The estimates are usually determined after considering claims development pattern, business exposure and various claims management actions. Apart from the usual Basic Chain Ladder (CL) method using paid and incurred claim triangles, Bornhuetter-Ferguson (BF) method (both paid and incurred), Ultimate Loss Ratio method and Frequency-severity approach are also considered. The final selection of IBNR provision for Private Car and Two-wheeler has been done based on a blended approach after comparing results from different methods. For Commercial Vehicles, ULR method has been used.

IBNR estimates (in INR ‘000s) from different methods as at 31st March 2022

LOB (Motor TP)	Paid CL	Incurred CL	Freq-Severity	ULR	Paid BF	Incurred BF	Selected
Private Car	55,00,000	58,00,000	98,05,500	-	1,25,00,500	1,15,15,500	1,00,50,000
Commercial Vehicle	70,00,000	75,10,250	-	1,35,00,000	1,20,25,000	1,15,00,500	1,35,00,500
Two-wheeler	90,250	2,45,500	2,15,000	-	2,78,250	3,05,125	3,00,000

- While the above methods give point estimates, we have also carried out stochastic reserving to estimate the mean and variance of the total ultimate claims arising from different accident year cohort.
- There has been some deterioration in claims reporting delay whereas the settlement lag has more or less remained consistent over the years.
- We have considered this phenomenon while selecting appropriate loss development factors in Chain Ladder method. Separate Margin for Adverse Deviation is applied to take care of any residual uncertainties.
- Reporting lag table for Motor TP liability:

Years	As at 31 st March 2020	As at 31 st March 2021	As at 31 st March 2022
1	62%	60%	55%
2	87%	86%	82%
3	95%	94%	92%
4	98%	97%	97%
5	99%	99%	98%

- Settlement lag table Motor TP liability:

Claims settled in	FY 2017-18	FY 2018-19	FY 2019-20	FY 2020-21	FY 2021-22
1 year	25%	24%	19%	20%	23%
1–3 years	31%	35%	29%	33%	33%
3-5 years	32%	35%	42%	35%	35%
> 5 years	12%	6%	10%	11%	9%

- The reserve utilisation table at the overall Company level is as follows:

Financial Year	Paid + Outstanding utilisation	Utilisation with closing IBNR
2015-16	62%	103%
2016-17	51%	96%
2017-18	59%	99%
2018-19	57%	98%
2019-20	39%	97%
2020-21	42%	100%
2021-22	50%	101%

Note:

$\text{Paid} + \text{Outstanding utilisation for FY 21-22} = (\text{Paid during FY 21-22 from March 21 outstanding claims} + \text{Paid during FY 21-22 from IBNR claims reported} + \text{March 22 outstanding from March 21 outstanding claims} + \text{March 22 outstanding claims from IBNR claims reported during FY 21-22}) / \text{Opening reserve for FY 21-22}$

$\text{Opening reserve for FY 21-22} = \text{March 21 closing outstanding claims} + \text{March 21 closing IBNR claims}$

$\text{Utilisation with closing IBNR} = (\text{Paid} + \text{Outstanding utilisation for FY 21-22}) + (\text{March 22 closing IBNR for prior accident years till March 21} / \text{Opening reserve for FY 21-22})$

For long-tailed line like Motor TP liability, IBNR will always be a significant component of total reserve as compared to short-tailed lines of business like Property, or Accident & Health. Due to the significant impact of Motor TP liability on total reserve, the similar trend is visible at overall level across the financial years.

Claim frequency:

While some improvement is observed in Motor TP liability claim frequency, it takes time for majority of claims being reported for any accident year, hence improvement of latest years needs to be monitored further before any suitable changes can be made to the IBNR estimation. Previous couple of years were affected by Covid induced lockdowns and hence the underlying frequency improvement may be somewhat different than what is seen in the table below.

Accident Year	Claim Frequency as at 31 st March 22	12 month claim frequency
2015-16	0.49%	0.21%
2016-17	0.47%	0.20%
2017-18	0.42%	0.18%
2018-19	0.39%	0.19%
2019-20	0.21%	0.09%
2020-21	0.16%	0.06%
2021-22	0.11%	0.11%

You have also collected some additional data points from Finance department as follows:

A) Geographical mix of Motor segment in different Companies during FY 2021-22

Company	Zone W	Zone X	Zone Y	Zone Z
Your Company	11%	17%	20%	52%
Competitor A	25%	30%	25%	20%
Competitor B	15%	33%	19%	33%
Competitor C	8%	30%	32%	30%
Competitor D	25%	28%	22%	25%

Severity from Industry reports:

Zone	Death Severity (In INR)	Injury Severity (in INR)
W	7,50,000	2,50,000
X	10,00,000	3,00,000
Y	9,00,000	2,75,000
Z	11,50,000	3,00,000

It is also known that the courts are usually more lenient towards claimant in Zone Z as compared to other regions.

B) While speaking with the CFO, you have been informed that the following information was previously shared by Company's Finance Team to the Director.

- Claim reserve can be broadly classified into two categories:
 - i. reserve for unexpired risk
 - ii. reserve for unpaid claims (outstanding plus IBNR)

First one is usually formula driven and carried out by Finance department while the second one is within Actuarial purview.

C) Comment from the Independent Actuarial Consultant

The Company also got the annual actuarial valuation peer reviewed. The Peer Reviewer has agreed with the appropriateness of the reserving but also stressed on the fact that the long-tailed Motor TP liability claims are affected by court award inflations, hence the Company needs to adequately factor the inflationary impact on claims all the time.

It was also recommended that the Company should keep a close look at the Survival Ratio before taking any decision on reserve release. Survival Ratio shows how long reserve will last before all outstanding and IBNR claims are paid if current paid or incurred claims development continues at a given rate.
